



Q METH 201 A: INTRODUCTION TO STATISTICAL METHODS

Class time: Mon/Wed: 1.10pm – 2.40pm

Location: PCAR 293

Instructor Name: Alfredo Effendy (*email:* aeffendy@uw.edu)

Office: Savery Hall (SAV) 319C

Office Hours:

- Wed: 2.50pm – 3.50pm (in-2person)
- Tu: 11.00am - 12.00pm (via [Zoom](#))
- **OR by appointment**

QUIZ SESSION

Teaching	Email	Section	Quiz Sessions	Location
Somin Park		AA	F: 10.50-11.50am	PCAR 293

OFFICE HOURS (*via Zoom OR in-person by appointment*)

Teaching	Section	Office Hours	Location
Somin Park	AA	Tu: 1pm -2pm	Zoom

Course Description: Probability and statistics are fundamental to business education. Business decisions are usually made with partial information in environments characterized by uncertainty and risk. The objectives of this course are to introduce you to (i) data analysis tools that are appropriate for generating useful information for decision making and (ii) a framework for analyzing decisions based on partial information. To achieve these objectives, we will develop a foundation of probability and statistics.

Topics include: representing data through numerical summaries and graph, measuring association between variables, measuring uncertainty using probability rules and probability distributions, estimating population parameters, testing hypotheses, and conducting regression analyses. We use examples to illustrate applications of the topics.

NOTES:

-  Attend the section for which you registered.
-  I begin classes on time and (usually) end classes on time, which means
-  Please ARRIVE BEFORE classes BEGIN
-  Please STAY UNTIL classes END

COURSE MATERIALS:

Course Website – Courses offered by the Foster School use the **Canvas** learning management system. Please check the course website SEVERAL times each WEEK.

Required textbook:

Anderson, David R., Dennis J. Sweeney, and Thomas A. Williams. *Essential of Modern Business Statistics with MS Office Excel*, (6th, 7th and 8th editions), Cengage Learning.

Software – I will post brief Canvas videos (under Canvas Modules) on how use MS Excel to analyze data and solve statistical problems similar to those used in class.

ACCOMMODATION:

DRS Accommodations. I assist all students who are registered with the Disability Resources for Students [DRS] office, 011 Mary Gates, 206.543.8924. Please **NOTE:** Each student registered with DRS needs to **meet with me** as soon as possible to discuss their specific accommodation.

RELIGIOUS ACCOMMODATIONS

Washington state law requires that UW develop a policy for accommodation of student absences or significant hardship due to reasons of faith or conscience, or for organized religious activities. The UW's policy, including more information about how to request accommodation, is available at [Religious Accommodations Policy \(https://registrar.washington.edu/staffandfaculty/religious-accommodations-policy/\)](https://registrar.washington.edu/staffandfaculty/religious-accommodations-policy/).

- Accommodations must be requested within the first two weeks of this course using the [Religious Accommodations Request form \(https://registrar.washington.edu/students/religious-accommodations-request/\)](https://registrar.washington.edu/students/religious-accommodations-request/).

FACE COVERING POLICY

Face coverings can be optionally worn in our classroom during class time.

ACADEMIC INTEGRITY/CODE OF CONDUCT:

I expect ALL students to follow the HIGHEST ethical standards when completing ALL assignments, and when interacting with other people associated with the course, especially other students and teaching assistants.

In addition, students are to complete ALL examinations and quizzes **without** assistance of any kind. Students may, however, work together to apply ideas and concepts on homework (HW) sets. Students who work together on HW sets MUST write their OWN final answers. The MINIMUM penalty for using unauthorized materials (including, but not limited to, other students' note sheets or/and too many note sheets) during exams and quizzes and for providing answers that are very similar, number-for-number or word-for-word, on any examination, quiz, or HWs will be ZERO credit.

Foster Undergraduate Code of Conduct:

As a student in QMETH 201 you acknowledge that you are a member of a learning community in the Foster School of Business that is committed to the highest academic standards. As a member of this community, you agree to uphold the standards of **honesty, respect, and integrity**, and accept the responsibility to encourage others to adhere to these standards.

Course Support: Because course material is cumulative, avoid falling behind. It is difficult to do well on the course if you don't understand the material as presented. If you do fall behind, please seek assistance.

Office Hours – In addition to my (*zoom*) office hours, TAs will hold 1 office hours per week to support student learning (TA's schedule will be announced on the 1st day).

IMPORTANT: I encourage you to regularly attend office hours and to proactively ask questions about any topic. Please don't fall behind, but if you do, make use of the office hours to catch up!

Lectures & Technology: In preparation for each class (and starting with Lecture 2), please print individual lecture slides posted under ***Pages*** on Canvas. If possible, please **abstain** from using **smart phones** (voice, text, picture) or other digital devices during the lectures.

What you SHOULD NOT do:

1. Forget, or write incorrect, personal identifying information on quizzes/exams
2. Skip classes or/and quiz sessions
3. Seldom/never visit the Canvas website
4. Take poor notes
5. Review only exercise sets when preparing for quizzes/exams
6. Construct insufficient note/formula sheets for quizzes/exams
7. Fail to ask questions or/and seek assistance
8. Fail to access materials
9. Fail to submit assignments or/and take quizzes/exams
10. Over-confidence (common among those took AP statistics) or under-confidence (common among students who believe they “cannot do math”)

CLASS FORMAT

Most of the sessions will use lectures to teach analytical methods. We will introduce these methods using in-class examples. You are highly encouraged to ask questions and/or make comments during class. Your input in class is an essential part of the learning process, both for you and for your classmates.

Grade Components:

Midterm Exam	29.1%	150
Final Exam (Cumulative)	31.1%	160
Homework Sets – 8 HWs	15.5%	80
Quiz Session – 4 Activity Sheets	5.8%	30
Quiz Session – 4 Quizzes	15.5%	80
TOTAL	100%	500

Grade Policies:

1. Students are guaranteed a final course grade per the *point cutoffs* indicated in the table below. This assignment of grades roughly follows normal distribution (i.e., bell curve). In course like ours, the median grade is set at 3.1, which approximately translates into a range of points between 410.1-420. This grading policy is **imposed by the university**.

Points	Grade	Points	Grade	Points	Grade
480.1-500	4	362.6-375	2.9	225.1-237.5	1.8
470.1-480	3.9	350.1-362.5	2.8	212.6-225	1.7
460.1-470	3.8	337.6-350	2.7	200.1-212.5	1.6
450.1-460	3.7	325.1-337.5	2.6	187.6-200	1.5
440.1-450	3.6	312.6-325	2.5	175.1-187.5	1.4
430.1-440	3.5	300.1-312.5	2.4	162.6-175	1.3
420.1-430	3.4	287.6-300	2.3	150.1-162.5	1.2
410.1-420	3.3	275.1-287.5	2.2	137.6-150	1.1
400.1-410	3.2	262.6-275	2.1	125.-137.5	1.0
387.6-400	3.1	250.1-262.5	2.0	<125	0.0
375.1-387.5	3.0	237.6-250	1.9		

2. Grades are NOT negotiable. I consider requests for “special favors” or “additional unearned points” at least unethical, perhaps equivalent to cheating.
3. I do NOT “round up” grades and there will be no extra credit assigned.

COURSE REQUIREMENTS:

Exams, homework sets, and quiz session activities. Attendance and participation are your responsibility; however, attending and participating in classes and quiz sessions should help you pass the course.

Smart phones are NOT allowed for exams and quizzes, so I recommend you acquire dependable basic calculator with **base e** – you are responsible for knowing how to use relevant functions (check *YouTube* tutorials). NOTE: programmable calculators (e.g., the several TI 83 and TI 84 models) are allowed (but UNNECESSARY).

- EXAMS** – One Midterm Exam (~29% of total grade) and the Final Exam (~31% of total grade). Students can use **ONE** formula/note sheets for the **Midterm Exam** and **TWO** formula/note sheets for the **Final exam**. Formula/note sheet guidelines: **STANDARD PAPER (8.5” x 11”), HAND-WRITTEN** (must be your handwriting), **BOTH SIDES**.
- HOMEWORK** – On average, you will submit **ONE** exercise set (~15.5% of the grade) each **WEEK** *to our Canvas site*. (Individual deadlines will be announced.) Exercise sets will be available on the COURSE WEBSITE approximately 1 week (*usually on Wednesday night*) before they are due.

Homework submission format:

(1) To obtain HW credit, please submit your completed **COVER SHEET** from **CANVAS**) – a document which contains ANSWERS ONLY. You must show ALL relevant calculations and background work on these worksheets.

- To obtain credit, please **submit (i.e., upload to Canvas) completed cover sheet**
- Do NOT submit the original questions.
- Please make sure your work NEAT and CLEAR – we will NOT grade exercise sets that are messy or difficult to follow.

C. QUIZ SESSION GROUP ACTIVITIES – During $\frac{1}{2}$ of the quiz sessions, you will work in TEAMS of 4-5 students to complete and submit an **ACTIVITY SHEET** (6% of total grade). Activity sheets will contain problems that extend course material and are based on datasets recorded in MS Excel. (*We will use Excel's Analysis ToolPak add-in to analyze data.*)

D. QUIZ SESSION QUIZZES – During remaining quiz sessions, you will take QUIZZES (~15.5% of total grade). Quizzes will cover several weeks of material. Students can use ONE formula/note sheet for EACH quiz (see the formula/note sheet guidelines for examinations above). We will return the exams and other assignments during Quiz Sessions.

MISSING QUIZ: students who expect to miss a quiz session due to University-sanctioned issues (e.g., illness) must contact me, NOT a TA, in ADVANCE to arrange an accommodation.

WEEKLY OVERVIEW

1	6/23 – Monday 1: Basic ideas, types of data	6/25 – Wednesday 2: Distributions
2	6/30 – Monday 3: Central Tendency & Measure of Spread	7/2 – Wednesday 4: Probability Relationships
3	7/7 – Monday 5: General Discrete & Binomial	7/9 – Wednesday 6: Poisson & Hypergeometric
4	7/14 – Monday 7: Normal Distribution	7/16 – Wednesday 8: Uniform & Exponential Distribution
5	7/21 – Monday MIDTERM EXAM	7/23 – Wednesday 9: Sampling & Sampling Distributions
6	7/28 – Monday 10: Confidence Intervals (<i>Part 1</i>)	7/30 – Wednesday 11: Confidence Intervals (<i>Part 2</i>)
7	8/4 – Monday 12: Hypothesis Testing (<i>Part 1</i>)	8/6 – Wednesday 13: Hypothesis Testing (<i>Part 2</i>)
8	8/11 – Monday 14: Covariance and Correlation	8/13 – Wednesday 15: Simple Linear Regression
9	8/18 – Monday 16: Final Exam Review	8/20 – Wednesday FINAL EXAM (@1.00 – 3.00pm in PCAR 293)

Date	TENTATIVE SCHEDULE
BASIC IDEAS, TYPES OF DATA	
6/23 – Mon	 Lecture Notes 1  <i>Chapter 1 (EMBS)</i>
DISTRIBUTIONS – TABLES, GRAPHS, AND SHAPE	
6/25 – Wed	 Lecture Notes 2  <i>Chapter 2 (EMBS)</i>
Quiz Session 1	
6/27 – Fri	➤ Group Activity 1
DISTRIBUTIONS – CENTRAL TENDENCY & STANDARD DEVIATION & VARIANCE	
6/30 – Mon	 Lecture Notes 3  <i>Chapter 3 (EMBS)</i>
PROBABILITY RELATIONSHIPS AND STATISTICAL INDEPENDENCE	
7/2 – Wed	 Lecture Notes 4  <i>Chapter 4 (EMBS)</i>  Homework 1
Independence Day (No Quiz Session)	
7/4 – Fri	➤ Take Home Quiz 1
DISCRETE DISTRIBUTIONS: GENERAL DISCRETE & BINOMIAL	
7/7 – Mon	 Lecture Notes 5  <i>Chapter 5 (EMBS)</i>
DISCRETE DISTRIBUTIONS: POISSON & HYPERGEOMETRIC	
7/9 – Wed	 Lecture Notes 6  <i>Chapter 5 (EMBS)</i>  Homework 2
Quiz Session 2	
7/11 – Fri	➤ Group Activity 2
CONTINUOUS DISTRIBUTIONS – NORMAL	
7/14 – Mon	 Lecture Notes 7  <i>Chapter 6 (EMBS)</i>
CONTINUOUS DISTRIBUTIONS – UNIFORM & EXPONENTIAL	
7/16 – Wed	 Lecture Notes 8  <i>Chapter 6 (EMBS)</i>  Homework 3
Quiz Session 3	
7/18 – Fri	➤ Quiz 2
SAMPLING AND SAMPLING -DISTRIBUTIONS	
7/23 – Wed	 Lecture Notes 9  <i>Chapter 7 (EMBS)</i>

7/21 – Mon	 Homework 4
MIDTERM EXAM	
7/21 – Mon	MIDTERM EXAM
Quiz Session 4	
7/25 – Fri	➤ Group Activity 3
CONFIDENCE INTERVALS (population proportion)	
7/28 – Mon	 Lecture Notes 10  <i>Chapter 8 (EMBS)</i>
CONFIDENCE INTERVALS (population mean)	
7/30 – Wed	 Lecture Notes 11  <i>Chapter 8 (EMBS)</i>  Homework 5
Quiz Session 5	
8/1 – Fri	➤ Quiz 3
HYPOTHESIS TESTING – ONE POP. MEAN	
8/4 – Mon	 Lecture Notes 12  <i>Chapter 9 (EMBS)</i>
HYPOTHESIS TESTING – ONE POP. PROPORTION	
8/6 – Wed	 Lecture Notes 13  <i>Chapter 9 (EMBS)</i>  Homework 6
Quiz Session 6	
8/8 – Fri	➤ Group Activity 4
CORRELATION AND COVARIANCE	
8/11 – Mon	 Lecture Notes 14  <i>Chapter 3 (EMBS)</i>
SIMPLE LINEAR REGRESSION	
8/13 – Wed	 Lecture Notes 15  <i>Chapter 14 (EMBS)</i>  Homework 7
Quiz Session 7	
8/15 – Fri	➤ Quiz 4
FINAL EXAM REVIEW	
8/18 – Mon	Final Exam REVIEW
8/19 – Tue	 Homework 8
FINAL EXAM	
8/20 – Wed	FINAL EXAM (PCAR 293) 1.00pm -3.00pm